



Course Code and Title: ITE104 – Information Management

Lesson 3: Week 3

Topic: System Development Life Cycle

Introduction

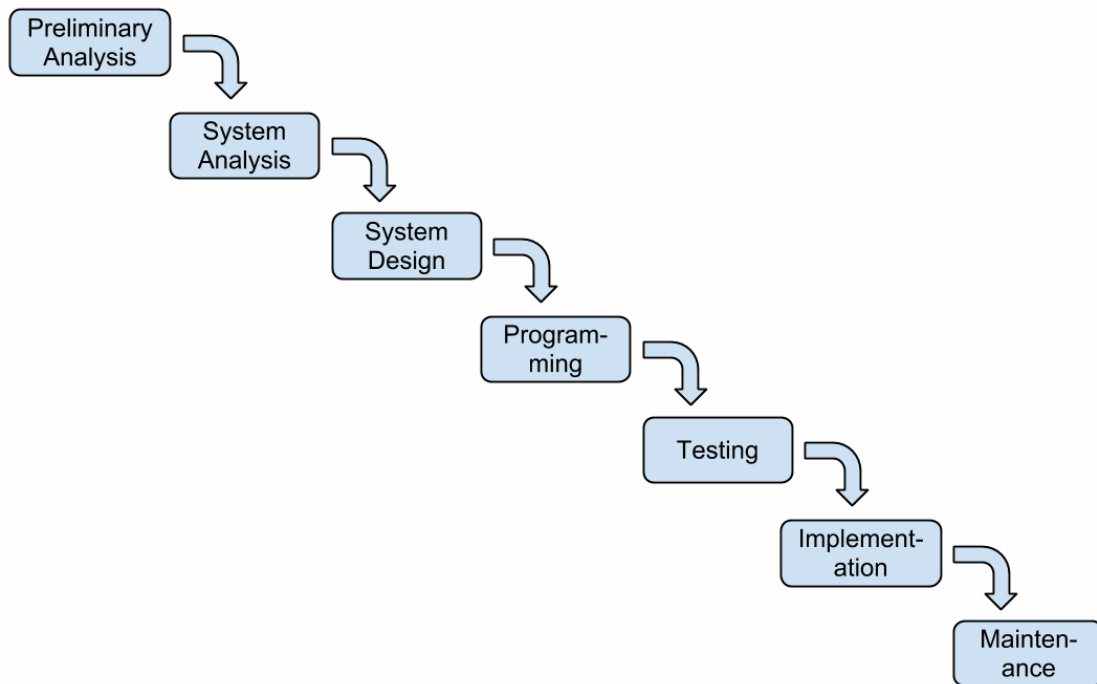
The systems development life cycle (SDLC), also referred to as the application development life-cycle, is a term used in systems engineering, information systems and software engineering to describe a process for planning, creating, testing, and deploying an information system. The systems development life-cycle concept applies to a range of hardware and software configurations, as a system can be composed of hardware only, software only, or a combination of both.

Lesson Presentation:

System Development Life Cycle

The first development methodology we are going to review is the systems-development life cycle (SDLC). This methodology was first developed in the 1960s to manage the large software projects associated with corporate systems running on mainframes. It is a very structured and risk-averse methodology designed to manage large projects that included multiple programmers and systems that would have a large impact on the organization.

Various definitions of the SDLC methodology exist, but most contain the following phases.



1. Preliminary Analysis. In this phase, a review is done of the request. Is creating a solution possible? What alternatives exist? What is currently being done about it? Is this project a good fit for our organization? A key part of this step is a feasibility analysis, which includes an analysis of the technical feasibility (is it possible to create this?), the economic feasibility (can we afford to do this?), and the legal feasibility (are we allowed to do this?). This step is important in determining if the project should even get started.

2. System Analysis. In this phase, one or more system analysts work with different stakeholder groups to determine the specific requirements for the new system. No programming is done in this step. Instead, procedures are documented, key players are interviewed, and data requirements are developed in order to get an overall picture of exactly what the system is supposed to do. The result of this phase is a system-requirements document.

3. System Design. In this phase, a designer takes the system-requirements



document created in the previous phase and develops the specific technical details required for the system. It is in this phase that the business requirements are translated into specific technical requirements. The design for the user interface, database, data inputs and outputs, and reporting are developed here. The result of this phase is a system-design document. This document will have everything a programmer will need to actually create the system.

4. Programming. The code finally gets written in the programming phase. Using the system-design document as a guide, a programmer (or team of programmers) develops the program. The result of this phase is an initial working program that meets the requirements laid out in the system-analysis phase and the design developed in the system-design phase.

5. Testing. In the testing phase, the software program developed in the previous phase is put through a series of structured tests. The first is a unit test, which tests individual parts of the code for errors or bugs. Next is a system test, where the different components of the system are tested to ensure that they work together properly. Finally, the user-acceptance test allows those that will be using the software to test the system to ensure that it meets their standards. Any bugs, errors, or problems found during testing are addressed and then tested again.

6. Implementation. Once the new system is developed and tested, it has to be implemented in the organization. This phase includes training the users, providing documentation, and conversion from any previous system to the new system. Implementation can take many forms, depending on the type of system, the number and type of users, and how urgent it is that the system become operational. These different forms of implementation are covered later in the chapter.

7. Maintenance. This final phase takes place once the implementation phase is complete. In this phase, the system has a structured support process in place: reported bugs are fixed and requests for new features are evaluated and implemented; system updates and backups are performed on a regular basis.



System Life Cycle – Stages and Activities

To appreciate fully the development of a system and its place within the organization, knowledge of the life cycle of systems is necessary. Like human beings, all systems go through a life cycle. Systems originate, develop, and, finally, decline. Several models have been developed that illustrate the concept of a system life cycle. In this section, four perspectives of system life cycles are discussed.

- General system life cycle
- Information system life cycle
- Discontinuity of information system life cycle
- Organization-wide information system life cycle

General system life cycle

Information systems are similar to biological, social, and political systems. All systems go through a system life cycle. All systems have a birth or development period, a period of growth, a period of maturity, and a period of decline or deterioration.

The following example demonstrates the four phases of the general system life cycle.

Birth and development (phase 1)

Ten years ago, the health information department of a community hospital implemented a system to track the release of information from patient records. The goal of the system was to allow health record employees to document the requests for information, the verification of those requests (as needed), and the description of the documents copied/released—to whom, why, when, and whether there was a charge associated with completing the request. The system was primarily a departmental system to meet the needs of this 300-bed facility.



Growth and maturity (phases 2 and 3)

In the intervening years, the hospital has formed several alliances that have broadened the scope of the services provided, and the laws related to release of patient-identifiable information have changed. Among the hospital changes is the expansion of outpatient services through acquisition of several clinics in the community, the addition of an outpatient surgery facility, and alliances with substance abuse, rehabilitation, and skilled nursing facilities. At first, the release-of-information tracking system was able to expand to keep pace with the institution's growing needs. However, as the hospital evolved into a health care enterprise, the needs for tracking and monitoring have changed. In addition, related laws have changed. For example, the Health Insurance Portability and Accountability Act and many state laws have specific regulations related to release of personally identifiable health information. In building the needed release-of-information tracking system, the organization will need to evaluate all the current and pertinent federal and state laws that apply to their particular organization. The HIM professional should take the lead in this endeavor.

Decline (phase 4)

The current release-of-information tracking system, which was designed primarily for internal departmental use, does not meet today's need for a more comprehensive, updated, enterprise-wide distributed system. The organization, or "enterprise," now has more components, perhaps with different rules related to release of information, for example, general health versus mental health or radiology images. The organization also has more "doors" through which information might be released. The formal release of information practices may be well understood, but clinical and administrative staff may not recognize disclosure of information that occurs through secure e-mail or staff who have access to electronic clinical reports and receive what appears to be a legitimate request for information. Tracking of release of information, including to whom, what was released, and for what purpose are part the formal HIM



department routine but may not occur if information is released through the many other doors that are now available in the organization or enterprise.

Tasks in the Development

Main System Development Life Cycle Phases

As we covered before, the SDLC is used as a conceptual model that includes the procedures and policies necessary to develop or alter a system throughout a life cycle. The end result should be a high-quality system that meets or exceeds customer expectations and is within time and budget constraints. Within this framework, these are the main system development life cycle phases:

Planning - Initial phase of the SDLC process that sets out to discover, identify, and define the scope of the project to decide the course of action and specifically address the issues that are going to be solved by the new system solution. This crucial phase sets the tone for the overall success of the project, which is why it is during this phase that thorough research is performed to determine resources, budget, personnel, technical aspects, and more.

Analysis - The purpose of this phase is to understand the business and processing needs of the information system project. Here, the development team considers the functional requirements of the system to assess how the solution will meet the end user's expectations. The end user requirements are determined and documented and a feasibility study is typically performed to determine if the project is feasible from a financial, organizational, social, and technological standpoint.

Design - After a comprehensive analysis phase, the design phase surfaces. In this third phase, the elements, components, security levels, modules, architecture, interfaces and data of the system are defined and designed to evaluate how the finished system will work and what it will look like. The system design is produced in detail to ensure the system will include the necessary features to meet all functional and operational aspects of the project.



Development - An approved design phase is the catalyst for the next and fourth phase: the development of the information system. In this phase, the development team is hard at work writing code and constructing and fine-tuning the technical and physical configurations necessary to build the overall information system. This is considered by many as the most robust phase in the life cycle as all the labor-intensive efforts are made here, signifying the real start of software production and installation of hardware as necessary.

Testing and integration - This phase involves the Quality Assurance (QA) team who is in charge of performing the overall system testing to determine if the system solution meets the set of business goals and if it performs as expected. Additionally, in this phase, all the different components and subsystems of the solution are brought together to bring the whole integrated system alive. As we've discussed in previous articles, testing is becoming increasingly important as it helps ensure customer satisfaction by establishing that the system is fault-free.

Implementation - After the system is given a green light from the QA team, it is brought into a production environment. In essence, during this phase, the project is released to be used and/or installed by end users.

Maintenance - In this final phase, end users can fine-tune the system as necessary to increase performance, add new features and capabilities, or meet new requirements brought to the table by the client. This phase ensures the system remains relevant and usable by replacing old hardware, improving the software, evaluating performance, and applying new updates to make sure it meets all necessary standards and includes the latest technologies to face cybersecurity threats.

Process Life Cycle

Information Processing Cycle - Categorizing and clustering data is an integral part of information processing. Information processing is a term used to describe the analysis of data, with the goal of extracting meaning from it. The main function of the



body's information processing system is to maintain homeostasis (homeostasis is defined as balanced environment, or a state of health). Often, when we have a new idea or insight, we don't stop and process it. We immediately get excited about the new thought and move on to the next thing. This is called the "information processing cycle."

What is Information Processing?

Information processing is the process of changing or converting information into meaningful information. It is processed, organized or classified data which is useful for the receiver. Information is the processed data which may be used "as it is" or may be put to use along with more data or information. The receiver of information takes actions and decisions based on the information received. Collected data must be processed to get meaning out of it, and this meaning is obtained in the form of information. Information processing is read as a part of other topics such as information processing theory, information visualization, information science, information technology, data science and statistics, etc. Further information is useful & meaningful only if it has these characteristics:

Timely – Information should be available when required, a delay in obtaining information renders it useless.

Accuracy – Accuracy of information has a significant impact on the decision-making. Possibilities of even slightest errors should be minimized.

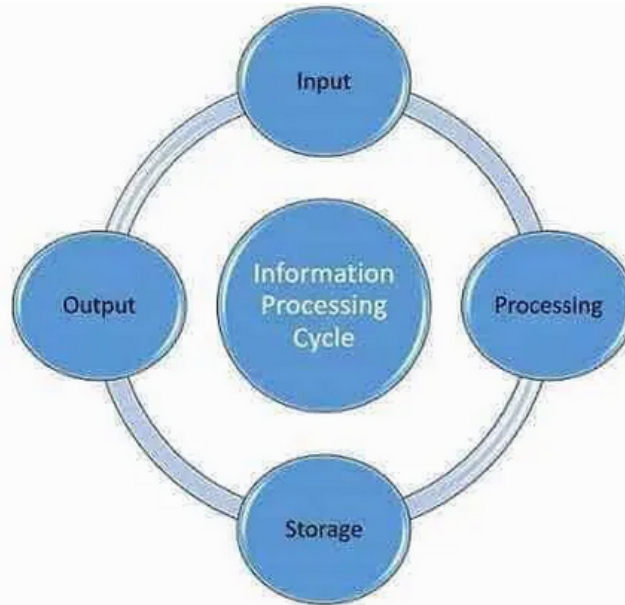
Completeness – Information should be complete. Incomplete information causes incorrect and unintended results.

Comprehensive – Information which is incomprehensible is useless for the receiver. This becomes a case of information failure as the sender sent the information, but it was of no use for the receiver, thus is not considered as "information."

Understanding Information Processing Cycle



Information processing cycle is a sequence of events consisting of Input, Processing, Storage & Output. To understand more about what is information processing cycle it is a good idea to compare it with the human information processing cycle. These events are similar as in case of the human information processing cycle. In the case of the computer, the computer has its own information processing cycle. Useful data are applied to the computer. Applying information to the computer and reprocess in the computer. This is followed by a similar process.



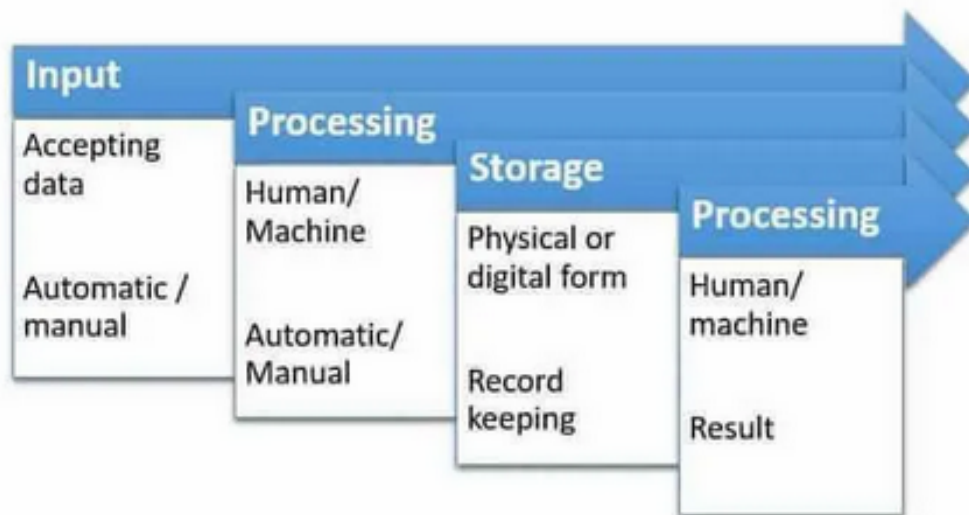
Four Phases of the information processing cycle

1. **Input:** Computer receives data and instructions.
2. **Process:** Computer applies instructions to data to produce information (organized



Data).

3. **Storage:** Saving the information for a subsequent use or use in future.
4. **Output:** Computer sends information to people in a usable format.



1. **INPUT** – Entering data into the computer

A. Feeding the collected raw data or data entry in the cycle for processing.

This is the raw data which is supplied for processing & obtaining information.

B. Input can be done by utilizing various input devices such as keyboards, mice, flatbed scanners, barcode readers, joysticks, digital data tablets (for graphics drawing), electronic cash registers, etc.

C. Input process is most essential as any error in the input data or the input process will result in wrong output.

2. **PROCESSING** – Performing operations on the data

A. Once the input is provided the raw data is processed by a suitable or selected processing method. Processing information is the most crucial step



as it allows the processed data in the form of output which will be used further.

B. Processing is usually done by CPU (Central Processing Unit) in a computer. CPU is the crucial component for getting the operations done. The speed of processing depends on the computing power and processing power of the processor.

C. Processing can be done by using various processing software's and computer system depending upon the requirement.

3. STORAGE – Saving data in a soft/physical form

A. The data which is to be fed and the output which will be generated needs to be stored. This stage is at times overlapping with the output stage since.

B. Storage can be done on various storage devices such as external hard disk, inbuilt hard disk, data centers, pen drives, micro SD cards, compact disks or even in registers.

4. OUTPUT – Results obtained, i.e., information

A. This is the outcome, and the raw data provided in the first stage is now “processed,” and the data is useful and provides information and no longer called Data. This might be further used for data visualization.

B. This can be used as it is or used for further processing along with more data. Output device can be a paper or in form of display screen such as monitors or phone screen.

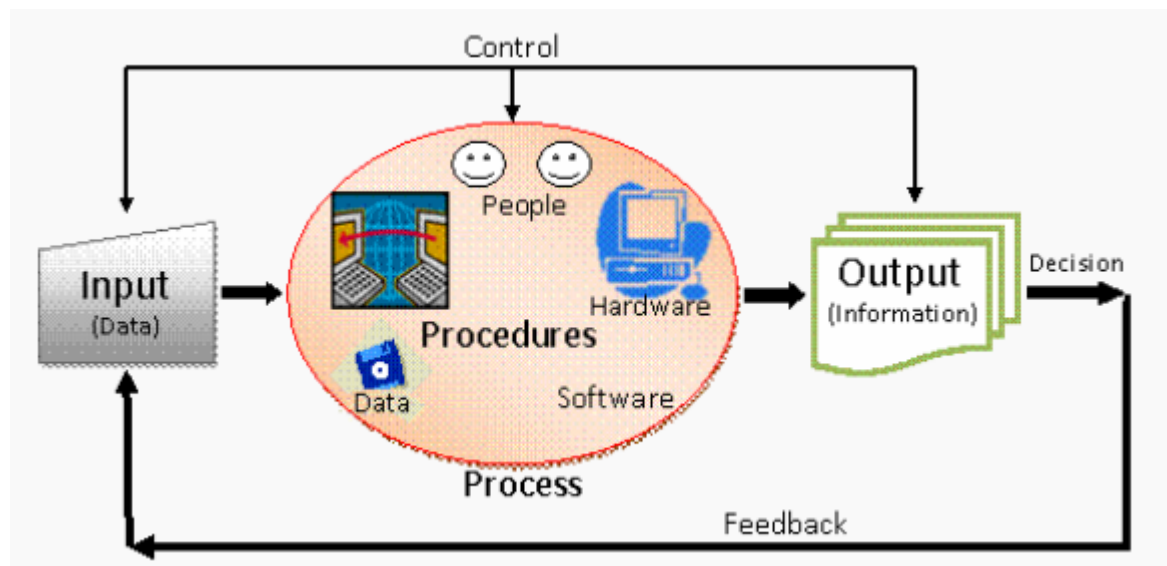
System Development Approaches

The word system is possibly the most overused word in our vocabulary and is used quite often in everyday life. Generally, we speak about political system, education system, human body system, social system, transportation system, banking system,



computer system etc. The common feature, which all systems share is that they are collection of elements integrated to achieve the required goals. The term system is derived from Greek word system, which means an organized relationship among functioning units or components. A system exists because it is designed to achieve one or more objectives. In present context, the word System can be defined as a group of interdependent components in an organized way, which works together to achieve one or more objectives as per specified plan. Interdependent components may refer to physical parts or managerial steps known as subsystem of a system. For example, a dairy plant can be considered as a system in which various departments and sections are joined together for a common goal.

A system may be regarded as a set of entities or elements interacting among themselves in a certain pre-specified manner to process inputs and produce outputs (objectives) in a defined time period. Output of a system may be quantitative or qualitative. The concept of a system is shown in fig. 5.1 as given below:



Characteristics of a System



As defined, a system is an organized relationship of various subsystems or components to achieve some objectives. For example, a computer system has a number of subsystems like Keyboard, Monitor, CPU, Mouse, etc. A Computer system is just an organized relationship of these subsystems in a planned way. These subsystems are also dependent of each other to achieve set targets. Thus the definition of a system suggests the following characteristics that are present in all systems.

Organization - This refers to a systematic order of components working together as per order to achieve centralized objective. It is the arrangement of subsystems in a specific structure and order that help to achieve objectives.

Interaction - This refers to the manner in which each subsystem functions, interacts or communicates with other components of the system. Special interfaces are designed to share data/ information among subcomponents.

Interdependence - It means that subsystems of a system are dependent on one another to achieve objectives of a system. This can be compared with a situation like a complex task which is divided into number of simple tasks (i.e. subsystems). Each simple task performs its work and produces output. Output of one task may become input of some other task. Like this a given complex problem is finally solved. Similarly in a system output of one subsystem may be an input of some other subsystem.

These subsystems are coordinated and linked together according to a plan.

Integration - This refers to holistic view of the system that how different subsystems are tied together to achieve central objective. A Bottom up approach is followed to integrate subsystems.

Planned approach - In order to achieve centralized objectives, a system should work according to a planned approach. The planned approach is a set of pre-laid policies, procedures, rules, budget, strategies and schedules to achieve goals.

Central objective - Each system is developed to achieve specified centralized



objectives. All subsystems are developed and integrated to achieve centralized objectives keeping the unique identity of each subsystem. Each subsystem works independently and interacts with other subsystems. Each subsystem contributes something directly or indirectly in achieving the system objectives.

Summary:

The main objective of this chapter is to gain knowledge about the working of the information system and the role of System Development Life Cycle (SDLC). It will integrate various aspects of the input & output of the Information System into SDLC. An information system is a combination of five elements human, data, software, hardware and network organized together to convert the given input into output by processing data into information or it is also said that information system is a system of organized attributes grouped and worked together as a single unit. The system resources used for processing information is software and hardware. Hardware refers to equipment' used for transforming information as equipment resource and software refers to collection of programs, as material resources.

Reinforcement:

Research (5 points Each) - Using MS word application, answer the following question. Save it as Activity1- (LastName, FirstName-Section).Docx and send to your Instructors Email-Gmail.

1. Which capabilities of the information systems have been employed in this application? How?
2. Suggest additional capabilities of information system that may be used to enhance the system.
3. What organizational effects would you expect from the new system?
4. Who was driving this project, MIS professionals or end-users?

Reference:

INFORMATION SYSTEM AND SYSTEM DEVELOPMENT LIFE CYCLE

https://www.researchgate.net/publication/290179000_Information_System_and_System_Development_Life_Cycle



Dr. Filemon C. Aguilar Memorial College of Las Piñas – IT CAMPUS

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m_Development_Life_Cycle

Information systems life cycle and project management

<https://nursekey.com/8-information-systems-life-cycle-and-project-management/>

Information Processing Cycle | Meaning

<https://planningtank.com/computer-applications/information-processing-cycle>